

AI Made Simple: 25-Lesson Learning Series

Master Curriculum & Production Blueprint for an Illustrated eBook and Cartoon Video Series

This master blueprint helps educators, curriculum designers, illustrators, scriptwriters, and video producers create one shared learning experience for school and college students ages approximately 12–22. Learners move from foundational AI ideas to practical, responsible beginner projects through 25 short, visual, age-appropriate lessons.

How to use this blueprint
Each lesson brief maps to one illustrated eBook chapter and one 3–5 minute cartoon video. Use it as a master plan for writing, illustration, storyboarding, production, and review—not as a fully scripted production.

Contents

1. Audience, outcomes, teaching principles, characters, and visual style guide
2. 25-lesson curriculum map
3. Detailed lesson briefs
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7. Quality, safety, and human review note

Review required before publication
AI outputs can be inaccurate, incomplete, biased, or misleading. A qualified human must review all educational content, sources, permissions, accessibility, and age appropriateness before publication or classroom use.

1. Audience, Outcomes, Teaching Principles, Characters, and Visual Style Guide

Audience

This series is designed for school and college learners approximately ages 12–22, with mixed prior experience. It can support classroom teaching, clubs, self-paced study, workshops, and blended learning. Examples should remain inclusive, neutral, and useful across different backgrounds and access levels.

Learning outcomes

By the end of the series, learners should be able to:

- Explain core AI concepts in plain language and distinguish capability from consciousness.
- Distinguish AI, machine learning, deep learning, natural language processing, generative AI, and tool-using systems.
- Describe data, models, tokens, prompts, embeddings, vector databases, retrieval-augmented generation, agents, and APIs at an introductory level.
- Evaluate accuracy, bias, privacy, copyright, safety, and the need for human review.
- Plan, test, present, and improve a beginner AI project while communicating what AI can and cannot do.

Teaching principles

- **Concrete before abstract** — begin with familiar situations, then introduce technical language.
- **One main idea per lesson** — keep explanations focused and build complexity gradually.
- **Visual explanation before jargon** — use diagrams, gestures, examples, and short definitions.
- **Capability with limits** — explain uncertainty, bias, privacy, and verification alongside benefits.
- **Active learning** — use recall questions, paper activities, small experiments, and reflection.

- **Human judgment** — model source checking, review, inclusive examples, and responsible decisions.

Recurring characters

Nova
A friendly rounded robot guide with a navy-and-teal body and a warm yellow accent. Nova is expressive but clearly non-human and non-conscious: a fictional teaching mascot, not a person or a sentient being.

Maya
A curious student who asks practical questions, connects ideas to everyday life, and helps the audience notice what is useful.

Jordan
A thoughtful student who tests ideas, checks assumptions, and spots edge cases or missing information.

Characters should represent different backgrounds without relying on stereotypes. Keep their proportions, clothing logic, expressions, and color accents consistent across the eBook and video series.

Visual style guide

Use a premium editorial education system: clean white and soft cream fields, navy #102A43, teal #138A8A, warm yellow #F4C95D as a restrained accent, charcoal text, simple line icons, clean geometric shapes, generous margins, subtle diagrams, and readable sans-serif typography.

Use restrained cartoon illustrations and consistent character proportions. Maintain high contrast, captions, transcript planning, and alt-text planning. Every visual should clarify a concept, relationship, process, or decision rather than decorate the page.

2. 25-Lesson Curriculum Map

MODULE 1: AI FOUNDATIONS

Purpose: Establish a clear, non-mystical foundation for understanding AI, machine learning, deep learning, and the role of data.

Progression: Learners move from the broad idea of AI to the more specific ways systems learn patterns and improve through feedback.

1. What Is Artificial Intelligence?

2. AI vs Traditional Programming
3. What Is Machine Learning?
4. What Is Deep Learning?
5. How AI Learns from Data

MODULE 2: DATA, MODELS & LANGUAGE

Purpose: Give learners a working vocabulary for datasets, models, neural networks, language technologies, and large language models.

Progression: Learners first examine the ingredients and evaluation of models, then connect those ideas to language and next-token prediction.

1. What Is Data? Features, Labels, and Datasets
2. Training, Testing, and Model Accuracy
3. What Are Neural Networks?
4. What Is Natural Language Processing (NLP)?
5. What Is a Large Language Model (LLM)?

MODULE 3: HOW GENERATIVE AI WORKS

Purpose: Explain the building blocks of text generation and teach learners to ask for useful, checkable outputs.

Progression: Learners move from tokens and generation loops to prompts, structured outputs, and the limitations that require verification.

1. What Is a Token?
2. How Does a Chatbot Generate Text?
3. Prompt Engineering Basics
4. Structured Outputs: Getting Reliable Answers
5. AI Hallucinations, Bias, and Limitations

MODULE 4: MODERN AI TOOLS

Purpose: Introduce the systems that help AI work with meaning-based search, outside knowledge, tools, and multi-step goals.

Progression: Learners build from representations to retrieval, then to supervised tool use and bounded agent-like workflows.

1. What Are AI Embeddings?
2. What Is a Vector Database?
3. What Is Retrieval-Augmented Generation (RAG)?
4. What Are AI Agents?
5. How AI Uses Tools and APIs

MODULE 5: BUILD, APPLY & USE AI RESPONSIBLY

Purpose: Connect concepts to real-world uses, beginner project planning, safety, privacy, copyright, and reflection.

Progression: Learners survey applications, design two conceptual projects, then complete a responsible project workflow from idea to presentation.

1. AI in Everyday Life and Different Industries
2. Build a Simple AI-Powered Study Assistant (conceptual beginner project)
3. Build a Mini FAQ Chatbot Using a Knowledge Source (beginner project)
4. AI Safety, Privacy, Copyright, and Responsible Use
5. Your First AI Project: Plan, Test, Present, and Improve

3. Detailed Lesson Briefs

Lesson 1 — What Is Artificial Intelligence?

A. Lesson title and module: What Is Artificial Intelligence? | Module 1: AI Foundations.

B. Learner-friendly objective: By the end, learners can describe AI as a system designed to perform tasks that often require human intelligence, without assuming that the system is conscious or human-like.

C. Key concepts: Artificial intelligence means systems designed to perform tasks such as recognizing patterns or working with language; pattern recognition means finding useful regularities in information; a model is a mathematical or computational pattern-maker; consciousness is not an automatic feature of AI.

D. Everyday analogy: A spam filter sorts messages using patterns in examples. It can be useful without having opinions, feelings, or awareness.

E. Visual/cartoon idea: Nova helps Maya and Jordan sort picture cards into “pattern task” and “not necessarily AI.” Show a simple two-column diagram with examples such as image

sorting, a calculator, a human conversation, and a timer.

F. Simple practical activity: In pairs, sort ten everyday examples into “AI-like pattern task” and “not necessarily AI,” then explain one uncertain choice. Safety note: avoid treating a human-like interface as evidence of consciousness.

G. Two quiz questions: 1) Multiple choice: Which is an AI-like pattern task? Answer: recognizing objects in images. 2) True/false: An AI system is automatically conscious because it uses language. Answer: False; language output does not prove consciousness.

H. Bridge to the next lesson: Once learners know what AI is meant to do, they can compare two ways of building systems: writing rules directly and learning patterns from examples.

Lesson 2 — AI vs Traditional Programming

A. Lesson title and module: AI vs Traditional Programming | Module 1: AI Foundations.

B. Learner-friendly objective: By the end, learners can contrast explicit rules written by people with patterns learned from examples.

C. Key concepts: Traditional programming uses instructions and rules written explicitly; machine learning uses examples to find patterns; a rule is a condition-action instruction; a spam filter may combine rules with learned patterns.

D. Everyday analogy: A recipe tells a cook each step directly, while learning to recognize a family recipe may involve seeing many examples and noticing recurring ingredients and methods.

E. Visual/cartoon idea: Nova gives Maya a rulebook for spam and gives Jordan a tray of labeled messages. Animate the contrast between a decision tree and a pattern-learning board; show a “rules versus examples” split screen.

F. Simple practical activity: Write three rules for sorting messages as “important” or “not important,” then discuss messages that break the rules. Safety note: use invented messages rather than personal email.

G. Two quiz questions: 1) Short answer: What does traditional programming provide directly? Answer guidance: explicit instructions or rules. 2) Multiple choice: What does machine learning use to find patterns? Answer: examples or data.

H. Bridge to the next lesson: The next lesson names the process by which a system uses examples to learn patterns and make predictions: machine learning.

Lesson 3 — What Is Machine Learning?

A. Lesson title and module: What Is Machine Learning? | Module 1: AI Foundations.

B. Learner-friendly objective: By the end, learners can explain machine learning as a way for systems to learn patterns from data and use those patterns to make predictions or decisions.

C. Key concepts: Machine learning finds patterns in data; training is the process of adjusting a model using examples; a trained model applies learned patterns to new input; prediction is an output based on the input and the model.

D. Everyday analogy: A learner practices identifying birds from many labeled pictures, then uses the observed features to make a best guess about a new picture. Practice examples are not the same as the final identification task.

E. Visual/cartoon idea: Nova places labeled examples on a conveyor belt while Maya watches a model form a pattern map. Jordan presents a new example and asks whether the prediction is certain; show the flow “examples → training → new input → prediction.”

F. Simple practical activity: Use a small spreadsheet of fictional study hours and quiz results to discuss what a model might predict. Safety note: do not use grades or identifiable student records.

G. Two quiz questions: 1) True/false: Training a model and using a trained model for a new prediction are the same step. Answer: False. 2) Short answer: What does a model learn from? Answer guidance: patterns in examples or data.

H. Bridge to the next lesson: Some machine-learning systems use many connected layers to process complex patterns. The next lesson introduces deep learning and explains what “deep” really means.

Lesson 4 — What Is Deep Learning?

A. Lesson title and module: What Is Deep Learning? | Module 1: AI Foundations.

B. Learner-friendly objective: By the end, learners can explain deep learning as machine learning that uses multi-layer neural networks, while recognizing that “deep” refers to layers rather than understanding or consciousness.

C. Key concepts: A neural network is a mathematical model made of connected processing units; a layer is one stage of transformation; deep learning uses multiple layers; image recognition can build from simple visual features to more complex patterns.

D. Everyday analogy: Imagine several teams checking a photo in sequence: one notices edges, another notices shapes, and another combines them into a possible object. The teams follow mathematical rules; they do not understand the photo as a person would.

E. Visual/cartoon idea: Nova projects an image of a bicycle through three transparent layers: lines, shapes, and object pattern. Add a labeled diagram showing input, hidden layers, and output, with a note that layers do not imply consciousness.

F. Simple practical activity: Give groups a simple drawing and ask them to describe three stages of visual processing from pixels to an object guess. Safety note: present the stages as a teaching simplification, not a literal description of every system.

G. Two quiz questions: 1) Multiple choice: In “deep learning,” what does deep mainly refer to? Answer: the number of layers. 2) True/false: A deep-learning model is a digital brain with human understanding. Answer: False.

H. Bridge to the next lesson: Layers need examples to adjust their patterns. The next lesson follows the loop of data, training, predictions, feedback, and improvement.

Lesson 5 — How AI Learns from Data

A. Lesson title and module: How AI Learns from Data | Module 1: AI Foundations.

B. Learner-friendly objective: By the end, learners can describe how data, training, predictions, feedback, and improvement interact, and explain why data quality matters.

C. Key concepts: Data provides examples; training adjusts a model; predictions apply learned patterns; feedback identifies useful or mistaken outputs; representation is how information is recorded; unwanted patterns can be learned when data is incomplete or skewed.

D. Everyday analogy: A coach reviews practice attempts, gives feedback, and helps an athlete adjust. If the practice examples leave out an important situation, improvement may not transfer to that situation.

E. Visual/cartoon idea: Animate a circular loop around Nova: data → training → prediction → feedback → improvement. Maya notices a missing example while Jordan points to a prediction that fails on an edge case; include a data-quality checklist diagram.

F. Simple practical activity: Create a paper classifier for fruit using color, size, and texture, then add an unusual fruit and discuss whether the rules still work. Safety note: label guesses as guesses and avoid collecting personal data.

G. Two quiz questions: 1) Short answer: Why can a model learn unwanted patterns? Answer guidance: the data may contain gaps, bias, errors, or misleading representations. 2) True/false: More data automatically guarantees a fair and accurate model. Answer: False.

H. Bridge to the next lesson: To understand how examples become usable inputs, learners next examine data, features, labels, and datasets in more detail.

Module 1 key idea
AI systems can recognize patterns and produce outputs without being conscious. Their performance depends on how the task, data, model, and feedback are designed and reviewed.

Lesson 6 — What Is Data? Features, Labels, and Datasets

A. Lesson title and module: What Is Data? Features, Labels, and Datasets | Module 2: Data, Models & Language.

B. Learner-friendly objective: By the end, learners can identify data, features, labels, and datasets in a simple classification example.

C. Key concepts: Data is recorded information; a dataset is an organized collection of examples; a feature is an observable input such as color or size; a label is the target category or answer; labeled data includes target answers, while unlabeled data does not.

D. Everyday analogy: A fruit table might record color, weight, and shape as features, with “apple” or “orange” as the label. A bowl of unclassified fruit is data without labels until someone identifies it.

E. Visual/cartoon idea: Maya and Jordan place illustrated fruit cards into a table while Nova highlights feature columns and a label column. Show a simple dataset grid with one labeled row and one unlabeled row.

F. Simple practical activity: Build a paper or spreadsheet dataset of fictional fruit using three features and one label. Safety note: use drawings or invented values rather than personal information.

G. Two quiz questions: 1) Multiple choice: Which is a feature in a fruit classifier? Answer: color. 2) Short answer: What is a label? Answer guidance: the category or target answer associated with an example.

H. Bridge to the next lesson: Once examples are organized, they must be divided and evaluated. The next lesson explains training data, test data, accuracy, and overfitting.

Lesson 7 — Training, Testing, and Model Accuracy

A. Lesson title and module: Training, Testing, and Model Accuracy | Module 2: Data, Models & Language.

B. Learner-friendly objective: By the end, learners can explain why a model should be tested on examples it did not memorize and describe overfitting in simple terms.

C. Key concepts: Training data helps a model learn; test data checks performance on new examples; accuracy is the share of evaluated answers that are correct; overfitting happens when a model memorizes training examples instead of learning a useful general pattern.

D. Everyday analogy: Studying the exact answers to one worksheet may produce a high score on that worksheet but not on a new version. A fair test uses unseen questions that measure transfer.

E. Visual/cartoon idea: Nova gives Maya a practice deck and Jordan a hidden test deck. Show a scorecard comparing “memorized examples” with “new examples,” plus a simple overfitting illustration of a line that twists around every training point.

F. Simple practical activity: Split a small paper classification set into practice and test cards. Predict the test score before revealing the cards, then discuss why the result may not generalize. Safety note: do not interpret a tiny classroom exercise as a real performance benchmark.

G. Two quiz questions: 1) True/false: Test data should be examples the model has not used to learn. Answer: True. 2) Short answer: What is overfitting? Answer guidance: memorizing training examples rather than learning a pattern that works on new examples.

H. Bridge to the next lesson: Many models learn through connected processing units. The next lesson introduces neurons, weights, layers, and activation as a simplified mathematical picture.

Lesson 8 — What Are Neural Networks?

A. Lesson title and module: What Are Neural Networks? | Module 2: Data, Models & Language.

B. Learner-friendly objective: By the end, learners can describe neurons, weights, layers, and activation as parts of a simplified mathematical model.

C. Key concepts: A neuron is a small computational unit; a weight represents how strongly an input contributes; a layer groups units at one stage; activation transforms a combined signal; a neural network is a mathematical model, not a digital brain.

D. Everyday analogy: A panel of reviewers gives different importance to clues, combines them, and passes a result to another panel. The panels calculate according to rules; they do not think or feel.

E. Visual/cartoon idea: Jordan feeds three clue cards into connected circles while Nova adjusts labeled weight sliders. Show a clean input-layer-output diagram with arrows and a small activation “on or off” illustration.

F. Simple practical activity: Use paper arrows and numerical weights to calculate a toy score for “is this fruit likely ripe?” Safety note: call the arithmetic a simplified demonstration, not a full neural-network implementation.

G. Two quiz questions: 1) Multiple choice: What does a weight represent? Answer: how strongly an input contributes in the simplified model. 2) True/false: A neural network is literally a human brain in a computer. Answer: False.

H. Bridge to the next lesson: Neural networks can process more than images. The next lesson applies pattern processing to human language through natural language processing.

Lesson 9 — What Is Natural Language Processing (NLP)?

A. Lesson title and module: What Is Natural Language Processing (NLP)? | Module 2: Data, Models & Language.

B. Learner-friendly objective: By the end, learners can define NLP and identify common language tasks while recognizing that language depends on context.

C. Key concepts: NLP is AI for working with human language; classification assigns categories; translation changes language while preserving meaning as well as possible; summarization condenses information; speech-related tasks may convert speech to text or text to speech; ambiguity means one phrase can have several interpretations.

D. Everyday analogy: A librarian sorts requests by meaning, but “book about bats” still needs context to tell whether the reader means animals or sports.

E. Visual/cartoon idea: Nova routes the same sentence through four stations: classify, translate, summarize, and transcribe. Maya asks about “bats,” while Jordan adds a context card; show a context-dependent meaning diagram.

F. Simple practical activity: Give groups an ambiguous sentence and ask them to list two possible meanings, then rewrite it to make one meaning clearer. Safety note: do not use private conversations or voices without permission.

G. Two quiz questions: 1) Multiple choice: Which is an NLP task? Answer: summarizing a paragraph. 2) Short answer: Why can language be difficult for AI? Answer guidance: meaning changes with context, ambiguity, tone, and missing information.

H. Bridge to the next lesson: One important class of language models works with very large text collections. The next lesson explains what a large language model predicts and what it does not automatically know.

Lesson 10 — What Is a Large Language Model (LLM)?

A. Lesson title and module: What Is a Large Language Model (LLM)? | Module 2: Data, Models & Language.

B. Learner-friendly objective: By the end, learners can explain that an LLM is trained on large amounts of text to predict likely next tokens and does not automatically know truth or understand like a person.

C. Key concepts: A language model estimates likely continuations; large refers to scale such as model parameters or training data, not intelligence in a human sense; a token is a text chunk; generated language can be fluent while still being incorrect or unsupported.

D. Everyday analogy: A skilled autocomplete system has seen many writing patterns and suggests what might come next. A plausible suggestion still needs checking because it may complete the sentence incorrectly.

E. Visual/cartoon idea: Nova displays a sentence with several possible next-token cards. Maya asks which one is true, and Jordan explains that likely wording is not the same as verified knowledge; show a probability-style choice without inventing numerical values.

F. Simple practical activity: Ask learners to complete sentence starters in several ways, then compare which continuation sounds likely versus which is factually supported. Safety note: do not enter personal, confidential, or identifying information into an AI tool.

G. Two quiz questions: 1) True/false: A fluent LLM response is automatically true. Answer: False. 2) Short answer: What does an LLM generally predict? Answer guidance: likely next tokens or continuations based on context.

H. Bridge to the next lesson: To understand next-token prediction, learners need to know what a model processes as a token. Lesson 11 breaks text into those smaller chunks.

> **Module 2 watch out**
> A model can produce polished language without possessing human understanding or a built-in truth checker. Teach learners to separate fluency, accuracy, and evidence.

Lesson 11 — What Is a Token?

A. Lesson title and module: What Is a Token? | Module 3: How Generative AI Works.

B. Learner-friendly objective: By the end, learners can explain that tokens are chunks of text processed by a language model and recognize why token counts matter.

C. Key concepts: A token may be a word, part of a word, punctuation mark, or space depending on the tokenizer; tokenization converts text into processable chunks; context length describes how much tokenized material a model can handle in one interaction; token counts are not always equal to word counts.

D. Everyday analogy: A sentence can be packed into tiles before being moved through a machine. Some tiles hold whole words, while others hold word parts or punctuation.

E. Visual/cartoon idea: Nova places a sentence on a conveyor belt and divides it into colorful tiles. Maya manually groups word chunks while Jordan points out punctuation; show a rough token strip and a note that real tokenizers may split differently.

F. Simple practical activity: Give learners sample words and sentences to split into rough chunks, then compare their choices and discuss why length limits and costs may depend on token counts. Safety note: use invented or public text only.

G. Two quiz questions: 1) Multiple choice: Which could be a token? Answer: punctuation or part of a word. 2) True/false: Every token is exactly one complete word. Answer: False.

H. Bridge to the next lesson: Once text is represented as tokens, a model can repeatedly estimate what token should come next. The next lesson visualizes that generation loop.

Lesson 12 — How Does a Chatbot Generate Text?

A. Lesson title and module: How Does a Chatbot Generate Text? | Module 3: How Generative AI Works.

B. Learner-friendly objective: By the end, learners can describe a simplified next-token generation loop and explain why different settings can change outputs.

C. Key concepts: The model reads context; it estimates likely next tokens; a selection step chooses one token; the loop repeats; sampling is a method for choosing among possible outputs; different settings can change variation without making randomness equal creativity or consciousness.

D. Everyday analogy: A sentence-building game offers several next tiles after each move. Choosing a different reasonable tile can lead to a different finished sentence.

E. Visual/cartoon idea: Nova, Maya, and Jordan build a sentence one tile at a time. A looping diagram shows “read context → estimate → select → repeat,” with two branches producing different but grammatical endings.

F. Simple practical activity: Use paper cards to create several possible next words for a sentence and let groups choose different paths. Safety note: treat outputs as examples for discussion, not as evidence of inner thoughts.

G. Two quiz questions: 1) Short answer: What repeats during generation? Answer guidance: reading the current context, estimating likely next tokens, selecting one, and continuing. 2) True/false: A different output proves that the model has a different mood. Answer: False.

H. Bridge to the next lesson: People can guide generation by specifying a goal, context, constraints, examples, and format. The next lesson turns those elements into practical prompt habits.

Lesson 13 — Prompt Engineering Basics

A. Lesson title and module: Prompt Engineering Basics | Module 3: How Generative AI Works.

B. Learner-friendly objective: By the end, learners can improve a vague prompt by adding a goal, context, constraints, examples, and a desired format.

C. Key concepts: A prompt is an instruction or input for an AI system; a goal states what to accomplish; context supplies relevant background; constraints define limits; examples

demonstrate the desired pattern; format describes how the answer should be organized.

D. Everyday analogy: Asking a classmate “help me with this” is less useful than saying what the task is, what has already been tried, what constraints apply, and what kind of help is wanted.

E. Visual/cartoon idea: Maya gives Nova a vague sticky note, then Jordan adds five labeled cards: goal, context, constraints, example, format. Show the prompt becoming clearer through a before-and-after storyboard.

F. Simple practical activity: Rewrite “Explain photosynthesis” into a prompt for a 13-year-old learner with a short analogy, three key terms, and two quiz questions. Safety note: never enter sensitive personal data, confidential records, or private classroom information.

G. Two quiz questions: 1) Multiple choice: Which addition makes a prompt more specific? Answer: a desired format and audience. 2) Short answer: Name two useful prompt elements. Answer guidance: goal, context, constraints, examples, or format.

H. Bridge to the next lesson: Clear prompts can also request outputs that are easier to check and reuse. Lesson 14 explores structured outputs such as tables, checklists, and labeled fields.

Lesson 14 — Structured Outputs: Getting Reliable Answers

A. Lesson title and module: Structured Outputs: Getting Reliable Answers | Module 3: How Generative AI Works.

B. Learner-friendly objective: By the end, learners can explain how schemas, tables, checklists, and labeled sections make generated answers easier to inspect without guaranteeing correctness.

C. Key concepts: A structured output follows an agreed format; a schema describes required fields; validation checks whether the format is complete and usable; a checklist makes missing items visible; correct structure does not guarantee correct content.

D. Everyday analogy: A form with required boxes helps a receptionist collect information consistently, but a neatly completed form can still contain a wrong phone number.

E. Visual/cartoon idea: Nova transforms a paragraph response into a labeled table. Jordan checks required fields while Maya spots a factual error inside the tidy structure; show “format check” and “content check” as two separate steps.

F. Simple practical activity: Design a four-field template for a book recommendation: title, audience, reason, and evidence. Ask an AI tool or partner to fill it, then verify every field against the source. Safety note: use public or teacher-approved material.

G. Two quiz questions: 1) True/false: A structured answer is automatically accurate. Answer: False. 2) Short answer: What does validation check? Answer guidance: whether required fields and formatting rules are present and usable.

H. Bridge to the next lesson: Even clear prompts and tidy formats cannot remove all model errors. The next lesson focuses on hallucinations, bias, limitations, and human review.

Lesson 15 — AI Hallucinations, Bias, and Limitations

A. Lesson title and module: AI Hallucinations, Bias, and Limitations | Module 3: How Generative AI Works.

B. Learner-friendly objective: By the end, learners can identify unsupported or biased outputs and apply basic source-checking and human-review habits.

C. Key concepts: A hallucination is a confident-sounding but unsupported or incorrect output; bias is systematic skew in data, design, or results; limitations include incomplete context and outdated knowledge; source checking compares claims with trustworthy evidence; human review adds judgment and accountability.

D. Everyday analogy: A confident speaker may misremember a detail and repeat it smoothly. Confidence in delivery does not prove that the detail is correct or fair.

E. Visual/cartoon idea: Nova presents a polished but incorrect fact card. Jordan holds a magnifying glass over the source while Maya asks who might be missing from the example; show a three-step graphic: pause, check, revise.

F. Simple practical activity: Compare two short AI-generated claims with a teacher-approved source and mark each as supported, unsupported, or unclear. Safety note: do not publish or grade the claims until a qualified human has reviewed them.

G. Two quiz questions: 1) Multiple choice: Which response is a hallucination? Answer: a confident claim with no support that conflicts with a reliable source. 2) True/false: Human review is unnecessary when wording sounds professional. Answer: False.

H. Bridge to the next lesson: Generative systems can also compare meaning across text by using numerical representations. The next module begins with embeddings, a stretch concept for meaning-based similarity.

Module 3 key idea
Better prompts and better formats can make outputs easier to use, but they do not replace evidence, source checking, or human judgment.

Lesson 16 — What Are AI Embeddings?

A. Lesson title and module: What Are AI Embeddings? | Module 4: Modern AI Tools.

B. Learner-friendly objective: By the end, learners can describe embeddings as numerical representations that capture useful relationships or similarity in meaning.

C. Key concepts: An embedding is a list of numbers representing an item; similarity compares how close representations are; meaning-based search looks beyond exact words; embeddings are useful representations, not perfect meanings or complete understanding; **stretch concept** — vector space is a mathematical space used to compare representations.

D. Everyday analogy: Imagine a map of ideas where related topics are placed near each other. The map can help find neighborhoods of meaning, but it is not the full meaning of every idea.

E. Visual/cartoon idea: Nova places cards for “raincoat,” “umbrella,” “sunhat,” and “pencil” on an idea map. Maya asks why some cards are closer, while Jordan notes that the map can still make mistakes; show dots and distance lines.

F. Simple practical activity: Create a paper map of ten school-related ideas and place similar ideas near one another, then compare groups’ maps. Safety note: explain that this human activity is an analogy, not a computed embedding.

G. Two quiz questions: 1) Multiple choice: What can embeddings help compare? Answer: relationships or similarity in represented items. 2) True/false: An embedding is a perfect definition of an idea. Answer: False.

H. Bridge to the next lesson: Embeddings become especially useful when many representations must be stored and searched. The next lesson introduces vector databases as an introductory stretch concept.

Lesson 17 — What Is a Vector Database?

A. Lesson title and module: What Is a Vector Database? | Module 4: Modern AI Tools.

B. Learner-friendly objective: By the end, learners can explain that a vector database stores embeddings and retrieves nearby or similar representations for search and retrieval tasks.

C. Key concepts: A vector is an ordered list of numbers; a vector database stores vectors with related information; similarity search finds nearby representations; metadata adds useful labels or filters; **stretch concept** — an index helps organize many vectors for faster retrieval.

D. Everyday analogy: A library shelf organized by topic can help locate related books even when the searcher does not know the exact title. A map of shelves can make the search faster, but the result still needs checking.

E. Visual/cartoon idea: Nova runs a query through a map-like library of idea cards. Jordan retrieves three nearby cards while Maya checks whether the best-looking card actually answers the question; show query, neighbors, and source labels.

F. Simple practical activity: Sort ten invented FAQ cards into topical groups, then retrieve the three cards closest to a new question and explain the choice. Safety note: use approved, non-sensitive content only.

G. Two quiz questions: 1) Short answer: What does a vector database store? Answer guidance: numerical representations called vectors or embeddings, often alongside source information. 2) True/false: Similarity retrieval guarantees that the retrieved answer is correct. Answer: False.

H. Bridge to the next lesson: Retrieval can supply relevant knowledge to a language model before it writes an answer. The next lesson combines retrieval and generation in a RAG workflow.

Lesson 18 — What Is Retrieval-Augmented Generation (RAG)?

A. Lesson title and module: What Is Retrieval-Augmented Generation (RAG) | Module 4: Modern AI Tools.

B. Learner-friendly objective: By the end, learners can describe RAG as retrieving relevant information from a knowledge source and providing it to a model before generation.

C. Key concepts: Retrieval finds relevant source material; augmentation supplies that material as context; generation produces a response; citations or source labels support checking; retrieval quality affects answer quality; **stretch concept** — chunking divides a source into searchable sections.

D. Everyday analogy: A student looks up approved class notes before answering a question, quotes the relevant passage, and checks that the answer matches the notes.

E. Visual/cartoon idea: Animate the flow “question → retrieve → provide context → answer → cite/check.” Nova fetches note cards, Maya asks the question, and Jordan checks whether the retrieved card is relevant; include a five-step pipeline diagram.

F. Simple practical activity: Build a paper RAG simulation using a small approved knowledge source: retrieve two relevant cards, draft an answer, and cite the cards used. Safety note: use teacher-approved material and never treat retrieval as a substitute for review.

G. Two quiz questions: 1) Multiple choice: What happens before generation in RAG? Answer: relevant information is retrieved and supplied as context. 2) Short answer: Why does retrieval quality matter? Answer guidance: irrelevant or missing context can lead to a weak or incorrect answer.

H. Bridge to the next lesson: Some systems can choose several steps toward a goal and use tools along the way. The next lesson introduces AI agents with clear boundaries and supervision.

Lesson 19 — What Are AI Agents?

A. Lesson title and module: What Are AI Agents? | Module 4: Modern AI Tools.

B. Learner-friendly objective: By the end, learners can define an AI agent as a system that can plan or choose steps toward a goal and sometimes use tools within rules and permissions.

C. Key concepts: An agent-like system receives a goal; planning selects possible steps; tools perform approved actions; permissions limit what the system may do; supervision checks decisions; “agent” describes a system role, not a conscious being; **stretch concept** — a loop can observe, decide, act, and check.

D. Everyday analogy: A supervised trip planner checks a destination, compares routes, and suggests steps, but a person approves bookings and handles unexpected situations.

E. Visual/cartoon idea: Jordan gives Nova a homework-planning goal. Nova proposes steps, Maya checks the rules, and a stop sign appears before any action requiring permission; show a bounded loop with human approval points.

F. Simple practical activity: Design a paper workflow for planning a study session with three steps and two approval points. Safety note: do not give an AI tool access to accounts, purchases, messages, or personal schedules during the activity.

G. Two quiz questions: 1) True/false: Calling a system an agent means it is conscious. Answer: False. 2) Short answer: Why are permissions important? Answer guidance: they limit actions and protect people, data, and resources.

H. Bridge to the next lesson: Agents often depend on tools that provide calculations, searches, calendar functions, or other controlled services. Lesson 20 explains tools and APIs.

Lesson 20 — How AI Uses Tools and APIs

A. Lesson title and module: How AI Uses Tools and APIs | Module 4: Modern AI Tools.

B. Learner-friendly objective: By the end, learners can explain that tools and APIs provide controlled ways for software to access functions or data outside a model.

C. Key concepts: A tool is an approved function such as a calculator or search service; an API is a defined way for software systems to request and receive data or actions; permissions set boundaries; validation checks inputs and outputs; safe design includes limits, logs, and human review; **stretch concept** — an API contract describes expected inputs and outputs.

D. Everyday analogy: A vending machine accepts a limited request, follows fixed rules, and returns an item. It cannot safely provide anything outside its menu or permissions.

E. Visual/cartoon idea: Nova sends a clearly labeled request to a calculator tool, receives a result, and shows it to Jordan for checking. Maya points to the permission gate; include an input → tool → result → validation diagram.

F. Simple practical activity: Create a paper tool card for a calculator, calendar, or approved search service, listing allowed inputs, disallowed inputs, and a validation step. Safety note: do not connect real accounts or share credentials.

G. Two quiz questions: 1) Multiple choice: Which is an example of a tool? Answer: a calculator used through a controlled function. 2) True/false: An API removes the need to validate results. Answer: False.

H. Bridge to the next lesson: Tools and models appear in many sectors, from education to transport. The next module surveys everyday and industry uses while weighing benefits and risks.

Module 4 watch out
Retrieval, tools, and agents can make a workflow more capable, but every added connection also creates new questions about permissions, source quality, errors, privacy, and human oversight.

Lesson 21 — AI in Everyday Life and Different Industries

A. Lesson title and module: AI in Everyday Life and Different Industries | Module 5: Build, Apply & Use AI Responsibly.

B. Learner-friendly objective: By the end, learners can identify possible benefits, risks, and human-review needs across several AI use cases.

C. Key concepts: Recommendation systems suggest items; accessibility tools can support communication or navigation; healthcare support tools may assist professionals without replacing them; transport, education, agriculture, finance, and creative work each involve different goals and risks; human oversight checks context and consequences.

D. Everyday analogy: A calculator can make arithmetic faster, but a person still decides whether the numbers were entered correctly and whether the result fits the real situation.

E. Visual/cartoon idea: Nova opens doors to illustrated scenes for education, farming, transport, healthcare support, and creative work. Maya identifies a benefit in each scene while Jordan adds a risk and a human-review checkpoint; use a balanced benefits-and-risks matrix.

F. Simple practical activity: In groups, choose one sector and complete a table with purpose, possible benefit, possible risk, affected people, and human review. Safety note: discuss general use cases without making medical, financial, or legal recommendations.

G. Two quiz questions: 1) Short answer: Why does human oversight matter across industries? Answer guidance: people understand context, check consequences, and remain accountable. 2) Multiple choice: Which is a risk to examine? Answer: inaccurate or biased output affecting a real decision.

H. Bridge to the next lesson: Learners now apply the shared concepts to a contained study-assistant idea using approved class notes and explicit verification steps.

Lesson 22 — Build a Simple AI-Powered Study Assistant

A. Lesson title and module: Build a Simple AI-Powered Study Assistant (conceptual beginner project) | Module 5: Build, Apply & Use AI Responsibly.

B. Learner-friendly objective: By the end, learners can design a conceptual study assistant that turns approved class notes into summaries, flashcards, and practice questions while checking results against the notes.

C. Key concepts: Scope defines what the assistant will and will not do; a knowledge source is the approved material it uses; output format makes results consistent; verification compares outputs with notes; limitations explain what the assistant may miss or misunderstand.

D. Everyday analogy: A study partner organizes a shared set of class notes into questions and summaries, but the learner checks the notes before studying from the result.

E. Visual/cartoon idea: Maya gives Nova a short approved note set. Nova creates a summary, flashcards, and questions; Jordan checks each item against the source. Show a five-step pipeline: scope → source → format → verify → reflect.

F. Simple practical activity: Define scope, provide a safe knowledge source, request a specified output format, verify every item against the notes, and write two limitations. No real deployment is required. Safety note: exclude personal records, private class information, and copyrighted material that is not approved for use.

G. Two quiz questions: 1) True/false: A study assistant should be used without checking its questions against the notes. Answer: False. 2) Short answer: Name two useful outputs. Answer guidance: summaries, flashcards, or practice questions.

H. Bridge to the next lesson: The study assistant uses one approved source and several output types. Next, learners design a mini FAQ chatbot that retrieves relevant entries and can say “I don’t know.”

Lesson 23 — Build a Mini FAQ Chatbot Using a Knowledge Source

A. Lesson title and module: Build a Mini FAQ Chatbot Using a Knowledge Source (beginner project) | Module 5: Build, Apply & Use AI Responsibly.

B. Learner-friendly objective: By the end, learners can outline a small FAQ chatbot that retrieves relevant approved information, answers within scope, and handles unknown questions safely.

C. Key concepts: An FAQ knowledge source contains organized questions and answers; retrieval selects relevant entries; an answer draft uses retrieved information; “I don’t know” behavior prevents unsupported guessing; edge-case testing checks unusual or out-of-scope questions; privacy review protects people and data.

D. Everyday analogy: A reception desk searches a binder of approved answers, gives the relevant page, and directs an unusual question to a person instead of inventing a response.

E. Visual/cartoon idea: Jordan organizes FAQ cards while Maya asks a matching question and then an out-of-scope question. Nova retrieves a card for the first and displays “I don’t know—please ask a person” for the second; show a retrieval-and-fallback flowchart.

F. Simple practical activity: Organize ten approved Q&A entries, test five direct questions and three edge cases, draft responses, and add a fallback message. Safety note: use fictional or public information and review every answer for accuracy and privacy.

G. Two quiz questions: 1) Multiple choice: What should a chatbot do when its knowledge source does not support an answer? Answer: say it does not know or direct the user to a human. 2) True/false: Edge-case testing is optional when the main examples work. Answer: False.

H. Bridge to the next lesson: Projects become responsible only when their information, permissions, and use are considered. Lesson 24 provides a practical checklist for safety, privacy, copyright, accessibility, and integrity.

Lesson 24 — AI Safety, Privacy, Copyright, and Responsible Use

A. Lesson title and module: AI Safety, Privacy, Copyright, and Responsible Use | Module 5: Build, Apply & Use AI Responsibly.

B. Learner-friendly objective: By the end, learners can apply a simple decision checklist before sharing data, using AI output, or publishing an AI-assisted work product.

C. Key concepts: Privacy protects personal or sensitive information; copyright concerns rights in creative works; attribution gives credit; academic integrity means representing work honestly; accessibility supports different needs; responsible use combines permission, verification, transparency, and human judgment.

D. Everyday analogy: Before sharing a photo or using someone’s idea, a careful person asks whether they have permission, whether the use is safe, whether credit is needed, and whether the result is fair.

E. Visual/cartoon idea: Nova, Maya, and Jordan walk through five checkpoint doors labeled “allowed,” “safe to share,” “verifiable,” “credit,” and “human review.” Use a decision checklist diagram with a stop-and-check moment before publication.

F. Simple practical activity: Apply this checklist to a fictional project: Is it allowed? Is it safe to share? Can I verify it? Do I need to credit a source? Does a human need to review it? Safety note: escalate uncertain cases to a teacher, institution, or rights holder rather than guessing.

G. Two quiz questions: 1) Short answer: Name two questions from the decision checklist. Answer guidance: any two of the five listed questions. 2) True/false: Adding AI to an assignment automatically makes it academically acceptable. Answer: False; rules and disclosure requirements still apply.

H. Bridge to the next lesson: The final lesson combines problem definition, testing, presentation, improvement, and responsible use into a complete beginner capstone workflow.

Lesson 25 — Your First AI Project: Plan, Test, Present, and Improve

A. Lesson title and module: Your First AI Project: Plan, Test, Present, and Improve | Module 5: Build, Apply & Use AI Responsibly.

B. Learner-friendly objective: By the end, learners can plan a beginner AI project, test it with examples, record errors and improvements, present what it can and cannot do, and reflect on responsible use.

C. Key concepts: A problem statement identifies a useful need; users and success criteria define who benefits and what “working” means; testing compares behavior with examples; an error log records failures and changes; presentation communicates scope and limitations; iteration improves a design through repeated review.

D. Everyday analogy: A school club plans an event by defining the audience, testing the schedule, recording what went wrong, revising the plan, and explaining the final choices to others.

E. Visual/cartoon idea: Maya sketches a project, Jordan tests edge cases, and Nova displays an iteration wheel: choose → define → sketch → test → record → improve → present. End with a gallery scene where each character names one limitation.

F. Simple practical activity: Choose a problem, define users and success criteria, sketch the solution, test it with examples, record errors and improvements, present the project, and reflect on responsible use. Safety note: keep the project small, use approved data, and obtain human review before sharing results.

G. Two quiz questions: 1) Multiple choice: Which belongs in an error log? Answer: what happened, why it matters, and what change was tried. 2) Short answer: What should a project presentation explain besides benefits? Answer guidance: scope, limitations, evidence, risks, and responsible-use decisions.

H. Bridge to the next lesson: This is the end of the shared 25-lesson sequence and the beginning of continued learning. Encourage learners to ask better questions, verify important claims, protect personal information, and improve projects through evidence and reflection.

4. Repeatable Production Templates

A. eBook chapter template

Chapter number and title: [Lesson number] — [Lesson title]

What you'll learn: [One or two measurable learner outcomes]

1-sentence hook: [A familiar question, surprising contrast, or practical problem]

Plain-language explanation: [Introduce the everyday term first, then add the technical term in parentheses when helpful.]

Key terms: [Three to five terms with short definitions; mark advanced terms as "stretch concept."]

Everyday analogy: [A memorable comparison that remains accurate and does not imply consciousness.]

Visual plan: [Illustration callout, diagram callout, character action, on-screen labels, and accessibility description.]

Mini activity: [Paper, discussion, spreadsheet, or beginner-tool activity with a safety or verification note.]

Check your understanding: [Question 1 with answer guidance] [Question 2 with answer guidance]

Watch out: [One misconception, limitation, uncertainty, privacy issue, or source-checking reminder.]

Bridge to next chapter: [One or two sentences explaining the conceptual connection.]

Accessibility checklist:

- [ ] Use a clear heading hierarchy and logical reading order.
- [ ] Check readable contrast and sufficient text size.
- [ ] Write useful alt text for meaningful illustrations and diagrams.
- [ ] Provide captions and transcripts where relevant.
- [ ] Review language for clarity, inclusion, and plain-language accessibility.

B. Cartoon video storyboard template — 3–5 minute episode

Timing can flex to fit the lesson while preserving clarity. Keep on-screen text short, readable, and synchronized with narration. Use Nova as a fictional, clearly non-conscious teaching mascot, and keep Maya and Jordan’s roles consistent.

Timing	Narration	Visual action	On-screen text	Character/dialogue notes	Sound or transition notes	Quiz/pause moment
0:00–0:15 hook	Ask a familiar question or show a useful problem.	Open with a clear action, prop, or visual surprise.	One short hook phrase.	Nova introduces the problem; Maya or Jordan reacts.	Brief sound cue; quick but readable transition.	Invite a prediction.
0:15–0:35 learning goal	State what learners will be able to do.	Show the goal as a simple card or path marker.	“By the end…” plus a short outcome.	Jordan clarifies the success target.	Settle into the lesson rhythm.	Ask learners to listen for one key term.
0:35–1:30 concept explanation	Define the plain-language idea and then the technical term.	Use Nova’s board, diagram, or object metaphor.	Three to five short labels.	Alternate explanation with a practical question.	Use gentle emphasis, not distracting effects.	Pause for a quick recall question.
1:30–2:20 analogy or character scene	Connect the concept to an accurate everyday analogy.	Stage the analogy with Nova, Maya, and Jordan.	One analogy label or comparison.	Maya asks the practical question; Jordan tests the edge case.	Transition using a visual match or repeated shape.	Ask which part of the analogy maps to the concept.
2:20–3:10 visual example or mini	Walk through a small example, activity, or process.	Show steps, cards, a simple spreadsheet, or a	Numbered steps or a compact flow.	Nova demonstrates; students narrate	Keep sound effects subordinate to speech.	Invite learners to identify the next step.

demons		paper		what		
3:10–3:45 pause-and-quiz	Ask two clear questions without trick wording.	Hold on a clean quiz card and allow thinking time.	Question and answer choices only.	Give Maya and Jordan different first guesses, then resolve.	Use a short pause cue.	Include the answer or answer guidance.
3:45–4:30 recap and limitation	Summarize the main idea and state one limitation or safety note.	Return to the opening visual and add a review/check mark.	Three recap words plus one “watch out.”	Jordan states what must be checked; Nova avoids overclaiming.	Use a calm recap transition.	Ask learners to name one verification step.
4:30–5:00 bridge and call to action	Connect to the next lesson and suggest one small action.	Move the characters toward the next lesson signpost.	Next lesson title and one action.	Nova closes; Maya and Jordan preview the next question.	End with a consistent series cue.	Prompt learners to try, discuss, or reflect safely.

5. Production Workflow

#	Step	Output	Quality gate
1	Research and fact-check	Approved source notes, definitions, examples, and risk notes.	Claims are supported, terminology is accurate, and uncertainty is identified.
2	Lesson brief approval	Signed-off brief with objective, concepts, activity, quiz, visuals, and bridge.	Educator or subject reviewer confirms scope, sequence, accuracy, and age appropriateness.
3	eBook chapter script	Chapter draft following the reusable eBook template.	Writing is concise, accessible, inclusive, and aligned with the approved brief.
4	Cartoon script and storyboard	Timed narration, visual actions, dialogue, on-screen text, and sound notes.	Every scene advances the concept; timing is realistic; limitations are visible.
5	Illustration and character asset creation	Nova, Maya, Jordan, diagrams, icons, props, and reusable scene assets.	Characters are consistent, age-appropriate, diverse without stereotypes, and visually legible.
6	eBook layout	Editable chapter pages with hierarchy, diagrams, captions, and callouts.	Spacing, contrast, reading order, labels, and page-break behavior are checked.
7	Voiceover recording	Clean narration, character dialogue, pronunciation notes, and take log.	Audio is understandable, paced for learners, and reviewed for factual wording.
8	Animation and editing	Final video edit with illustrations, motion, voice, music, and transitions.	Visuals match narration, on-screen text is readable, and no unsupported claim remains.

9	Accessibility checks	Captions, transcript, alt text, readable contrast, audio checks, and visual-text review.	Captions are complete and synchronized; meaningful visuals are described; audio and text are usable.
10	Educational, factual, sensitivity, and rights review	Review log with corrections, permissions, source credits, and approval status.	Qualified human reviewers approve content, age fit, inclusivity, rights, privacy, and safety.
11	Publish and archive source files	Approved eBook, video, captions, metadata, editable files, exports, and archive record.	Human review is complete before publishing; final links, version, date, and owner are recorded.

Human review checkpoint
Do not publish or use an episode with learners until educational, factual, sensitivity, rights, privacy, accessibility, and age-appropriateness reviews are complete.

6. Episode Publishing Checklist and File Naming

Episode publishing checklist

- [ ] Lesson number and title match across the eBook, video, thumbnail, captions, and metadata.
- [ ] The learning objective is present and aligned with the lesson brief.
- [ ] Key terms are defined accurately in plain language.
- [ ] Quiz questions and answer guidance are checked.
- [ ] Claims and examples are fact-checked against approved sources.
- [ ] Visuals match the narration and are age-appropriate.
- [ ] Nova, Maya, and Jordan are visually and behaviorally consistent.
- [ ] Captions and transcript are complete and synchronized.
- [ ] Visual text passes readability and contrast checks.
- [ ] Music, voices, images, and fonts have the necessary rights.
- [ ] Privacy-sensitive information is excluded or appropriately protected.
- [ ] A qualified human reviewer has approved the final package.

- [ ] Source/editable files and export files are archived.
- [ ] Final links, version, date, and owner are recorded.

File naming convention

Use this consistent pattern: `AIS_L{lesson-number}_{short-topic}_{asset-type}_v{major.minor}_{YYYY-MM-DD}`

Part meanings: AIS identifies the series; L identifies the lesson number; short-topic provides a readable topic label; asset-type identifies the eBook, video, captions, thumbnail, or other deliverable; v{major.minor} records the revision; YYYY-MM-DD records the file date.

Examples:

- `AIS_L13_PromptBasics_ebook_v1.0_2026-10-29`
- `AIS_L13_PromptBasics_video_v1.0_2026-10-29`
- `AIS_L13_PromptBasics_captions_v1.0_2026-10-29`
- `AIS_L13_PromptBasics_thumbnail_v1.0_2026-10-29`

The example date is illustrative only and does not imply a production deadline. Record the actual date used for each approved or exported file.

7. Quality, Safety, and Human Review Note

Human review required before publication
AI-generated explanations, examples, scripts, illustrations, and quiz answers can be inaccurate, incomplete, biased, or misleading. A qualified human must review educational content, sources, permissions, accessibility, and age appropriateness before anything is published or used with learners.
Learners should question outputs, verify important claims, protect personal information, respect copyright, and use AI as a tool rather than an authority.

Series-wide quality commitments

- Use plain-language definitions first, followed by technical terms when they help learners progress.
- Describe Nova as a fictional teaching mascot and never imply that Nova or any AI system is conscious, sentient, or human.

- Show benefits and limitations together, with source checking and human review built into activities and production.
- Protect privacy, respect copyright and attribution, support accessibility, and keep examples inclusive and age-appropriate.
- Preserve the exact 25-lesson sequence across the eBook, video series, metadata, captions, and archive.

Master curriculum blueprint | Editable working document | Human review required before publication